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	PROCEDURAL MANUAL	
	RISK MANAGEMENT (CONSTRUCTION)	

OBJECTIVE	This procedure outlines the process for identifying, analyzing, evaluating, treating, monitoring, and reviewing risks during the construction phase of projects undertaken by BHBI. The aim is to minimize potential negative impacts on project cost, schedule, quality, and safety during construction.
SCOPE	This procedure applies specifically to the **construction phase** of projects, starting from the issuance of the Notice to Proceed (NTP) and continuing through to substantial completion and project handover. It does not cover pre-construction phases like design or procurement, although risks identified during those phases that could impact construction should be considered.

1 Risk Identification (Construction Phase)

- **Methods:**
  - **Site Inspections:** Regular site inspections by the Project Engineer, Site Engineer, and Safety Officer to identify potential hazards and risks.
  - **Construction Checklists:** Utilize construction-specific checklists to identify common risks associated with different construction activities (e.g., excavation, concrete work, steel erection, etc.). (See Appendix A for example checklists).
  - **Subcontractor Input:** Require subcontractors to identify and report risks associated with their scope of work.
  - **Safety Meetings:** Regular safety meetings (e.g., toolbox talks) to discuss potential hazards and risks.
  - **Incident Reports:** Analyze incident reports and near misses to identify underlying risks.
  - **Lessons Learned:** Review lessons learned from previous projects to identify potential risks on current projects.
- **Focus Areas:** During the construction phase, pay particular attention to risks related to:
  - **Site Conditions:** Unforeseen ground conditions, existing utilities, adjacent structures.
  - **Construction Activities:** Excavation, working at heights, heavy lifting, hot work, etc.
  - **Safety Hazards:** Falls, struck-by, caught-in/between, electrocution, etc.
  - **Weather:** Adverse weather conditions that could impact safety or progress.
  - **Material and Equipment:** Defective materials, equipment breakdowns, delivery delays.
  - **Subcontractor Performance:** Quality of work, adherence to schedule, safety compliance.
  - **Labor:** Labor shortages, skill gaps, labor disputes.
- **Documentation:** All identified risks are recorded in the Construction Risk Register (See Appendix B for a template).
  - **Deliverables:** Construction Risk Register

5.2 Risk Analysis (Construction Phase)

- **Likelihood:** Assess the probability of each construction risk occurring, considering factors like the frequency of the activity, site conditions, and controls in place. Use a scale (e.g., 1-5, where 1=Very Low, 5=Very High).
- **Impact:** Assess the potential consequences if the construction risk occurs. Consider impacts on project cost, schedule, quality, safety, and the environment. Use a scale (e.g., 1-5, where 1=Insignificant, 5=Catastrophic).
- **Documentation:** Record the likelihood and impact ratings for each risk in the Construction Risk Register.

5.3 Risk Evaluation (Construction Phase)

- **Risk Matrix:** Utilize a risk matrix (See Appendix C for an example) to determine the overall risk level (e.g., Low, Medium, High, Extreme) based on the likelihood and impact ratings.
- **Prioritization:** Prioritize construction risks based on their risk level. Focus on High and Extreme risks first.
- **Documentation:** Record the risk level and priority in the Construction Risk Register.

5.4 Risk Treatment (Construction Phase)

- **Risk Response Strategies:**
  - **Avoid:** If possible, eliminate the risk altogether during construction (e.g., change construction methods).
  - **Mitigate:** Implement measures to reduce the likelihood or impact of the risk during construction (e.g., additional safety measures, closer supervision, more frequent inspections).
  - **Transfer:** Contractually transfer the risk to another party, such as a subcontractor or insurance company (e.g., requiring subcontractors to carry specific insurance).
  - **Accept:** Acknowledge the risk and develop contingency plans for the construction phase (e.g., have backup equipment available, allocate contingency time in the schedule).
- **Action Planning:** Develop detailed action plans for each construction risk, including:
  - **Specific actions** to be taken during construction.
  - **Responsible parties** (e.g., Project Engineer, Site Engineer, Subcontractor, Safety Officer).
  - **Timeline** for implementing the actions.
  - **Required resources**.
- **Documentation:** Record the chosen risk response strategy and detailed action plan in the Construction Risk Register.
  - **Deliverables:** Construction Risk Treatment Plans or Construction Risk Mitigation Plans

5.5 Risk Monitoring and Review (Construction Phase)

- **Regular Monitoring:**
  - **Project Engineer and Site Engineer:** Continuously monitor construction activities, site conditions, and subcontractor performance to identify any new or emerging risks.
  - **Safety Officer:** Regularly monitors safety performance, conducts site safety inspections, and ensures compliance with safety procedures.
  - **QA/QC Personnel:** Regularly monitor the quality of construction work through inspections and testing.
  - Monitor the effectiveness of implemented risk responses.
  - Track the progress of action plans.
- **Formal Reviews:**
  - Conduct formal risk reviews at regular intervals during the construction phase (e.g., weekly or bi-weekly).
  - Involve the Project Manager, Project Engineer, Site Engineer, Safety Officer, QA/QC Personnel, and key subcontractors in the reviews.

- **Documentation:** Update the Construction Risk Register with the status of each risk, including any changes in likelihood, impact, or risk level.

5.6 Risk Communication and Reporting (Construction Phase)

- **Internal Communication:**
  - Regularly communicate construction risk information to the project team, including subcontractors.
  - Discuss risks in site meetings, progress meetings, and safety meetings.
- **External Communication:**
  - Communicate relevant risk information to the client, consultants, and other external stakeholders as required.
- **Reporting:**
  - **Project Engineer:** Reports on construction risks to the Project Manager.
  - **Project Manager:** Includes updates on construction risk management in regular project reports to senior management and the client.
  - **Deliverables:** Construction Risk Reports

6. Training

- All construction personnel (including subcontractors) will receive training on this Construction Risk Management Procedure.
- Specific training will be provided on hazard identification, risk assessment, and risk reporting.

7. Continuous Improvement

- This procedure will be reviewed and updated periodically to ensure its continued effectiveness and relevance to construction operations.
- Lessons learned from construction risk management activities will be incorporated into the procedure and shared across the company.

Appendix A: Example Construction Risk Checklists

Here are a few examples of checklists you can include in Appendix A:

Excavation and Trenching Checklist:

- Are all underground utilities located and marked before excavation begins?
- Is there a competent person designated for excavation safety?
- Are soil conditions assessed and classified?
- Is sloping, shoring, or shielding used where required?
- Are excavated materials stored at least 2 feet from the edge of the excavation?
- Are ladders or other safe means of access and egress provided?
- Are excavations inspected daily before work begins and after any rainstorm or other hazard-increasing occurrence?
- Is there a standby person designated when workers are inside deep excavations?

Working at Heights Checklist:

- Are guardrails, safety nets, or personal fall arrest systems (PFAS) in place where required?
- Are all personnel working at heights properly trained in fall protection?
- Are ladders secured and inspected before use?
- Are scaffolds properly erected, inspected, and tagged?
- Are openings and holes on elevated surfaces properly covered or guarded?
- Is there a rescue plan in place for workers using PFAS?

Concrete Work Checklist:

- Is formwork properly designed, erected, and inspected before concrete placement?
- Is rebar placement inspected and approved before concrete placement?
- Is the correct concrete mix design being used?
- Are concrete deliveries properly scheduled to ensure a continuous pour?
- Is concrete vibrated properly during placement?
- Are proper curing methods being used?
- Are concrete tests (slump, cylinders/beams) being performed according to specifications?

Appendix B: Construction Risk Register Template

Here's a template you can use for your Construction Risk Register. You can adapt it to fit your specific needs:

Risk ID	Project Phase	Risk Description	Likelihood (1-5)	Impact (1-5)	Risk Level	Risk Owner	Mitigation Actions	Contingency Plan	Status	Due Date
CR-001	Excavation	Collapse of trench due to unstable soil				Project Engineer			Open	
CR-002	Concrete	Late delivery of concrete affecting pour schedule				Site Engineer			Open	
CR-003	Steel Erection	Fall from height during steel erection				Safety Officer			Open	

Appendix C: Risk Matrix Example

You can use the same Risk Matrix example provided in the general Risk Management Procedure or adjust it to reflect the specific risk tolerance levels during the construction phase.

Likelihood	Impact	1 - Insignificant	2 - Minor	3 - Moderate	4 - Major	5 - Catastrophic
5 - Very High		Medium	High	High	Extreme	Extreme
4 - High		Medium	Medium	High	High	Extreme
3 - Moderate		Low	Medium	Medium	High	High
2 - Low		Low	Low	Medium	Medium	High
1 - Very Low		Low	Low	Low	Medium	Medium

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